PROJECT PROGRESS REPORT

(Progress Report no. 3)

1. Student Information

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Semester	3
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2. Project Details

Type	Research Project
Department/Domain	MCA CA(DS/ML) / AI & ML
Mentor Name	Srikant Singh
Start Date	08-06-2026
End Date	03-10-2026

3. Work Progress Summary

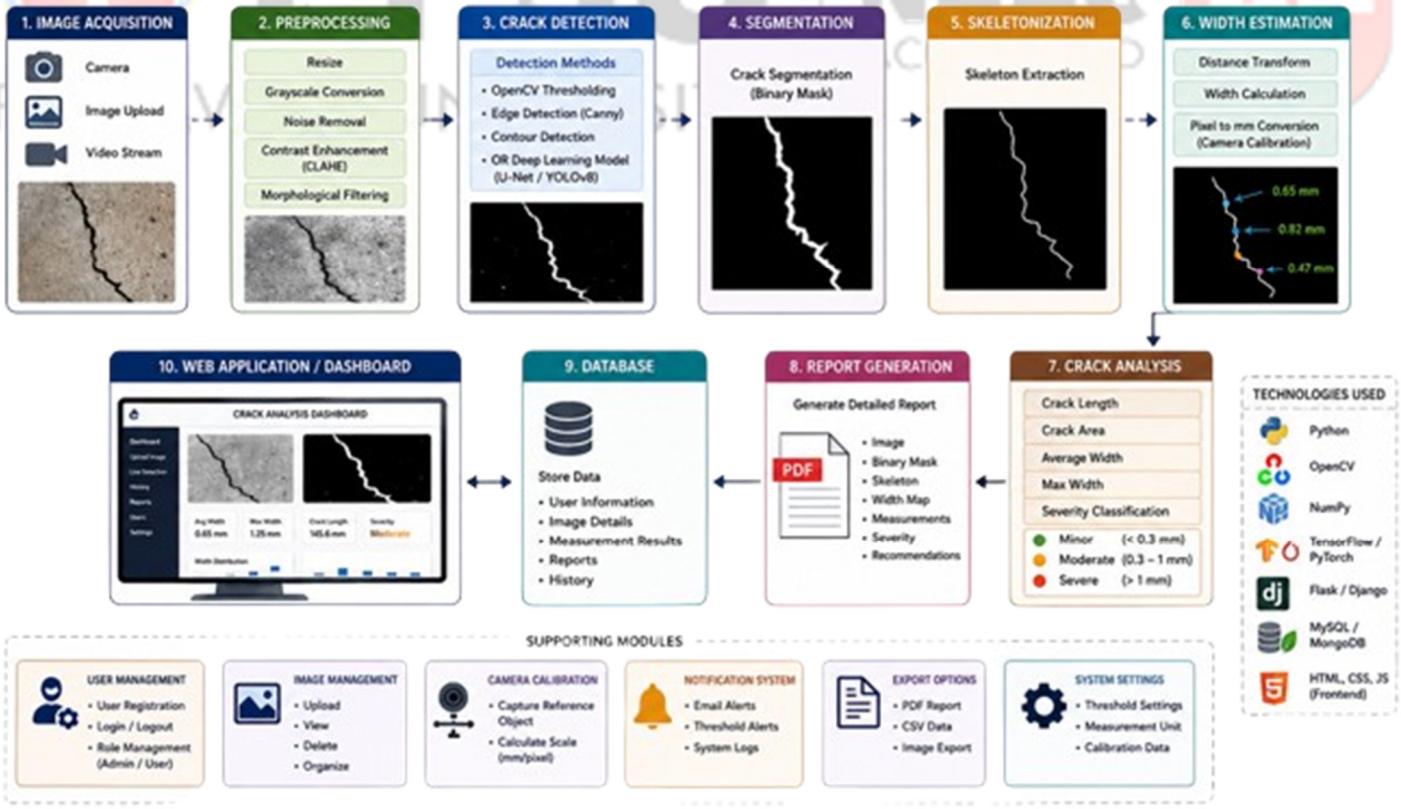
a) Project Title:

- OpenCV - Based System for Crack Identification and Width Estimation

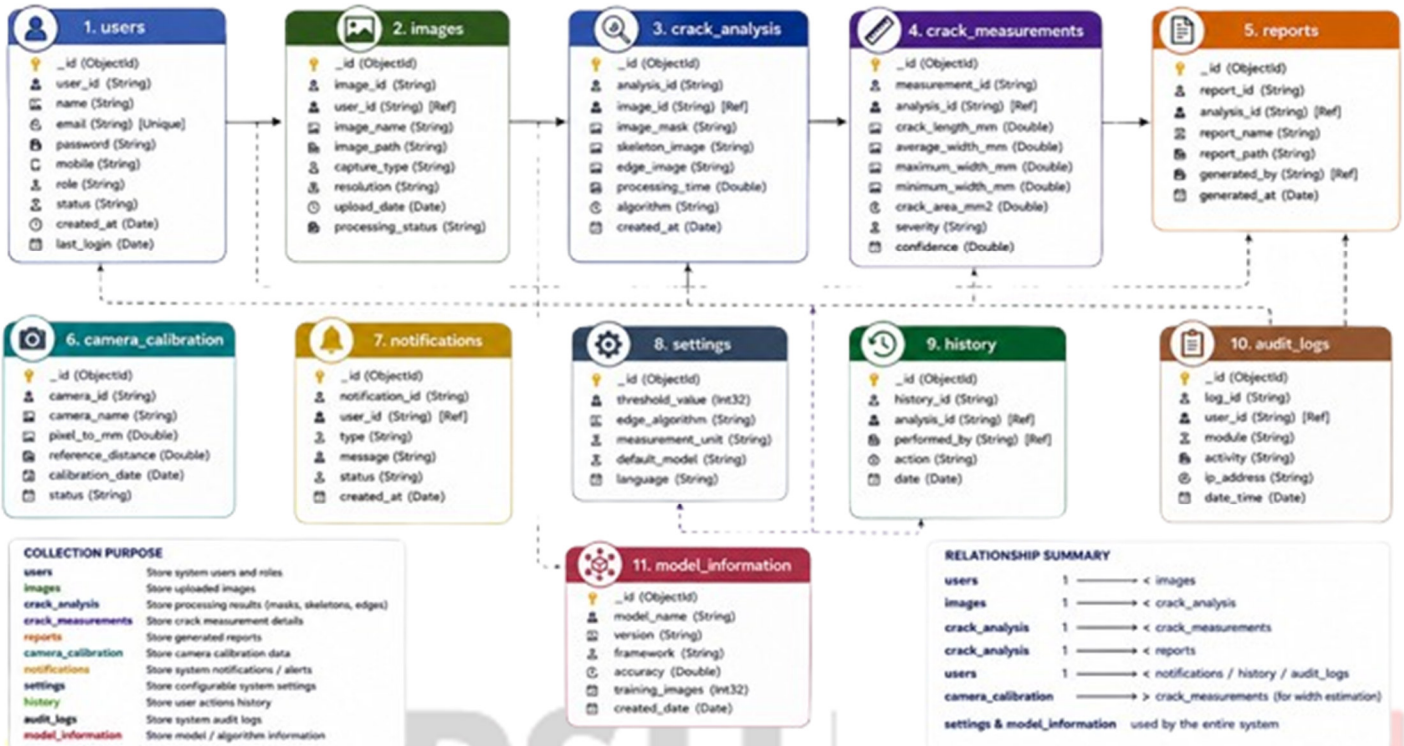
b) System architecture:

- The proposed system follows a modular architecture for automated crack detection and width estimation.
- The workflow begins with image acquisition using a camera or drone. The captured image is passed to the preprocessing module, where resizing, grayscale conversion, and noise reduction are performed.
- The enhanced image is then processed using OpenCV algorithms for crack detection through edge detection and contour extraction.
- The detected crack is analyzed by the width estimation module, which calculates crack dimensions using pixel measurements and calibration techniques.
- Finally, the crack is classified according to its severity, and the results are displayed and stored for future analysis.

Architecture Flow:



c) Database design:



d) Module Planning:

Module 1: Image Acquisition

Purpose: Acquire crack images from different sources.

Functions:

- Capture images using a camera.
- Upload images from the local system.
- Validate image format (JPG, PNG, BMP).
- Store image metadata.

Input: Crack image/video

Module 2: Image Preprocessing

Purpose: Improve image quality for accurate crack detection.

Functions:

- Resize image.
- Convert RGB to grayscale.
- Apply Gaussian Blur for noise removal.
- Enhance contrast using CLAHE.
- Perform normalization.

Input: Original image

Module 3: Crack Detection

Purpose: Detect crack regions from the preprocessed image.

Functions:

- Apply thresholding.
- Detect edges using Canny Edge Detection.
- Extract contours.
- Identify crack pixels.

Input: Enhanced image

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Module 4: Crack Segmentation

Purpose: Separate the crack from the background.

Functions:

- Generate binary mask.
- Remove unwanted objects.
- Refine crack boundaries.

Input: Detected crack

Module 5: Skeletonization

Purpose: Reduce the crack to a one-pixel-wide centerline.

Functions:

- Extract crack skeleton.
- Preserve crack topology.
- Prepare for width estimation.

Input: Binary mask

Module 6: Crack Width Estimation

Purpose: Calculate crack width accurately.

Functions:

- Apply Distance Transform.
- Measure crack width at every skeleton point.
- Convert pixels to millimeters using calibration.
- Compute minimum, maximum, and average width.

Input: Skeleton image

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Module 7: Crack Analysis

Purpose: Analyze crack characteristics.

Functions:

- Calculate crack length.
- Calculate crack area.
- Calculate average and maximum width.
- Generate statistical measurements.

Input: Width measurements

Module 8: Severity Classification

Purpose: Classify crack severity.

Functions:

- Minor (< 0.30 mm)
- Moderate (0.30–1.00 mm)
- Severe (> 1.00 mm)

Input: Crack width

Module 9: Report Generation

Purpose: Generate inspection reports.

Functions:

- Create PDF reports.
- Export CSV/Excel reports.
- Save processed images.
- Store measurement results.

Input: Analysis results

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Module 10: Dashboard / Web Application

Purpose: Provide a user-friendly interface.

Functions:

- User login.
- Upload crack images.
- View analysis results.
- Download reports.
- Search history.
- Manage users and settings.

Input: User request

4. Tasks Accomplished

- Conducted a comprehensive literature review on crack detection and width estimation techniques using OpenCV and Deep Learning.
- Finalized the project objectives, scope, and system requirements.
- Designed the overall system architecture, including image acquisition, preprocessing, crack detection, segmentation, skeletonization, width estimation, crack analysis, report generation, database, and dashboard modules.
- Selected the technology stack, including Python, OpenCV, TensorFlow/ PyTorch (optional), Flask/Django, MongoDB, NumPy, and Matplotlib.
- Designed the MongoDB database schema for storing user information, crack images, analysis results, reports, camera calibration data, and system logs.
- Planned and documented all software modules with their inputs, outputs, and functionalities.
- Set up the complete development environment by installing Python, OpenCV, MongoDB, Visual Studio Code, and the required libraries.
- Collected and organized crack image datasets for system development and testing.
- Developed the initial image preprocessing module, including image resizing, grayscale conversion, Gaussian filtering, contrast enhancement (CLAHE), and thresholding.
- Implemented the crack detection pipeline using OpenCV techniques such as edge detection, contour extraction, and binary segmentation.
- Designed the crack width estimation methodology using skeletonization and distance transform for accurate measurement.

5. Tasks in Progress / Next Plan

- initial module implementation
- coding approach

6. Learnings / Skills Acquired

- Gained a strong understanding of Computer Vision and Digital Image Processing concepts for crack identification and measurement.
- Learned to use the OpenCV library for image preprocessing, edge detection, contour extraction, segmentation, and feature extraction.
- Improved Python programming skills by implementing image processing algorithms and developing modular application components.
- Acquired practical knowledge of image enhancement techniques such as grayscale conversion, Gaussian filtering, histogram equalization (CLAHE), thresholding, and morphological operations.
- Learned the implementation of Skeletonization and Distance Transform algorithms for accurate crack width estimation.
- Developed an understanding of camera calibration and pixel-to-millimeter conversion for real-world crack measurement.
- Gained experience in MongoDB database design, including creating collections for users, images, crack analysis, measurements, reports, and system history.
- Learned to design a modular software architecture, workflow diagrams, and database schema for a scalable computer vision application.
- Enhanced knowledge of Flask/Django-based web application development for integrating image processing with a dashboard interface.
- Learned to generate structured outputs such as inspection reports, statistical summaries, and visualization dashboards from processed crack data.
- Strengthened problem-solving, debugging, analytical thinking, and software development practices while implementing and testing different project modules.

References:

- Sandbhor, S., Desai, P., Ranjan, A., Sharma, D., Baig, S., Agarwal, K., & Wadhwa, R. S. (2024, June). Computer Vision and Sensor-Based Concrete Crack Detection System. In *International Conference on Data Analytics & Management* (pp. 319-336). Singapore: Springer Nature Singapore.
- Giulietti, N., Revel, G. M., & Chiariotti, P. (2025). Automated vision-based concrete crack measurement system. *Measurement*, *242*, 115858.
- Kim, H., Lee, S., Ahn, E., Shin, M., & Sim, S. H. (2021). Crack identification method for concrete structures considering angle of view using RGB-D camera-based sensor fusion. *Structural Health Monitoring*, *20*(2), 500-512.
- Jin, S., Lee, S. E., & Hong, J. W. (2020). A vision-based approach for autonomous crack width measurement with flexible kernel. *Automation in Construction*, *110*, 103019.p

Evaluation (To be filled by Mentor)

Criteria	Bloom's Level	Excellent (20)	Good (15)	Average (10)	Poor (5)	Marks Obtained (Out of 20)
Problem Definition	L4–Analyze	Clearly defined, justified real-world problem	Defined but lacks depth	Vague problem	No clear problem	
Project Title Relevance	L2–Understand	Highly precise and aligned	Mostly relevant	Partially relevant	Misleading	
Scope of Work	L5–Evaluate	Well-defined scope, deliverables, constraints	Defined with minor gaps	Broad/unclear	No scope	
Related Study	L4–Analyze	3–5 sources analyzed critically	Sources listed	Minimal references	No study	
Clarity & Presentation	L3–Apply	Structured, professional	Minor issues	Some confusion	Poor	
Total Marks						

Evaluation

Student Signature	Mentor (Company) Signature

Mentor (Institute) Remarks

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